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Analyzing TF-IDF and BERT Approach for Bangla Text Classification Using Transformer-Based Embedding for Newspaper Sentiment Classification

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Abstract. Text classification classifies texts into relevant classes or categories, allowing users to find specific documents on the internet by searching with the keyword of that targeted class. Both traditional feature-based and transformer based Bert model for text classifications in the languages have become a very interesting field in Natural Language Processing . Newspapers are the most common source of current events news and information, and online newspaper portals are becoming increasingly popular as people become more internet dependent. This research aims to classify the Bangla newspaper articles into corresponding classes. Along with text classification, a comparison between the traditional Term Frequency-Inverse Document Frequency (TF-IDF) based and Bidirectional Encoder Representations from Transformers (BERT) based model for feature extraction will be brought up in this research. Machine learning algorithms such as Decision Tree, Random Forest Classifier, Support Vector Machine Classifier, and Logistic Regression Model will be implemented for classifying the newspaper articles. A couple of scenarios will be presented and contrasted based on the most accurate prediction and the maximum accuracy among all instances is 87.96%, which is derived from the logistic regression model and for the transformer-based feature extractor model, hence the BERT-based model showed superior to the conventional feature extractor methodology.

Keywords: Text Classification, BERT, TF-IDF, NLP, Category, Machine Learning Algorithms, Feature Extraction.

1 Introduction

Text is a combination of words that contain related meanings and are used by individuals to convey views, thoughts, viewpoints, and emotions. There are about seven thousand languages worldwide, with almost half of these having written forms. Text classification is a technique based on machine learning to classify or categorize texts in any language. Natural Language Processing (NLP) is an emerging topic of Artificial Intelligence that helps automate repetitive operations in human language. Text classification has made it simpler for users to search for and access websites and apps, making it more accurate and relevant [1]. Online platforms are more popular for news publication, making text categorization an effective method for discovering specific information or news. This research emphasizes on categorizing Bangla newspaper articles into their appropriate groups by evaluating ten thousand pieces of data from a dataset. It also focuses on performing a comparative analysis of the BERT-based feature extraction approach and the TF-IDF-based feature extraction method for targeted texts. This study comprises pre-processing the raw dataset, feature extraction, performance evaluation using four machine learning models, and feature tweaking with a transformer-based model. Text categorization is a developing focus in natural language processing, with BERT and TF-IDF being prominent methods for feature generation. In this research, two opposite feature extractor has been used for analyzing the better fitted model for news classification. This research aims at classifying sentiment from newspaper in order to understand the pattern of interest of the readers.

2 LITERATURE REVIEW

In [2], The authors did a study on Bangla text categorization using a multilingual transformer model, BERT and XLM-RoBERTa. They worked on sentiment analysis, news classification, and authorship attribution. They evaluated three datasets and attained an accuracy rate of 71.7%, 74.4%, and 76.0% for sentiment analysis. For news categorization, they attained accuracy rates of 91.28%, 92.70%, and 93.41% for the BERT-base, XLM-RoBERTa-base, and XLM-RoBERTa-large models. They also employed pre-trained models for emotion detection and authorship attribution, generating considerable results. In [3], The author offered a new integrated model of TF, IDF, and ICF. They utilized their own created dataset that includes 4000 Bangla documents. The documents were of multiple classes, in a total of eight classes. They employed normal methods like tokenization, deleting extraneous letters and punctuation, etc. The authors employed three feature extractor techniques which are TF, TF-IDF-ICF, and TF-IDF. After feature creation, they employed Naïve Bayes as the classifier. The classifier obtains accuracies of 91.76%, 98.87%, and 93.32% correspondingly. The hybrid model TF-IDF-ICF performs the best for categorizing the Bangla papers into their respective categories. In [4], the author worked on text categorization for the Indian language that covered the Bangla language also. This research proposes combining Inverse

Class Frequency (ICF) with the term frequency (TF) and inverse document frequency (IDF) approaches can offer higher performance for feature development for the Bangla language. After feature development and reduction of dimensionalities, they applied classifiers as such Decision Tree (J48), Naive Bayes, K-Nearest Neighbor, Naive Bayes Multinomial, Multilayer Perceptron, and PART. Among these, MLP had the most accurate findings. Before dimension reduction, the accuracy was 97.65%, while after dimension reduction, the accuracy was 98.03%. This research reveals that reduction strategies combined with the combination of three feature extraction methods, the performance becomes significantly better. The authors of [5] performed the Bangla blogs-based text classification. It was conducted by employing nine distinct supervised learning methods including a decision tree, support vector machine, multilayer perceptron, etc., and three feature extractors such as unigram TF-IDF, bigram TF-IDF, and trigram TF-IDF. Staring with pre-processing the dataset, tokenization was done, and tokens were input into the TF-IDF procedures indicated above. Then the models were trained and the classification algorithms on them. The suggested model has approximately 4000 blogs to be sorted among 8 categories. The support vector machine with unigram TF-IDF achieved the best accuracy which is 87.4%.

3 METHODOLOGY

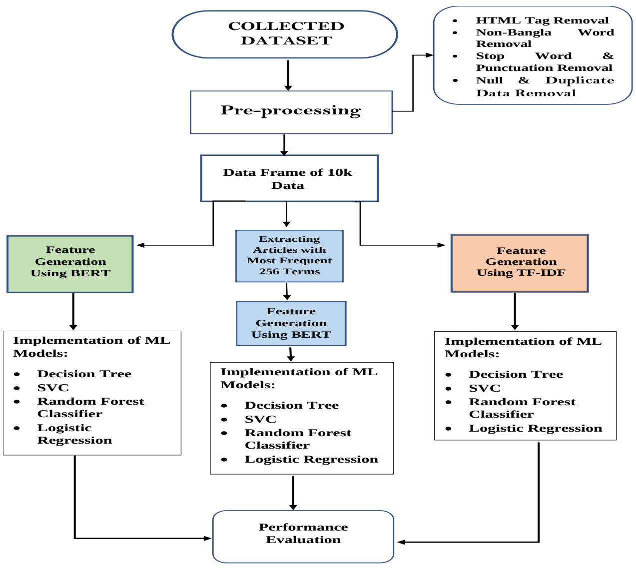


Fig. 1. Proposed Performace Evaluation Block Diagram

The proposed approach for the research such as the consecutive steps to complete the research that is to classify the targeted newspaper articles will be discussed in this section. This research is mainly empirical as the results of this research have been analyzed based on empirical cases. Analyzing the experimental results of the experimental cases, the conclusion has been drawn and so, this empirical research has been conducted by some consecutive procedures. This section will give an overview of the chosen dataset, and how it has been extracted and preprocessed to undergo the final stage which is the implementation of the machine learning models to get the classification report and accuracy score of the predicted classes. In this research, text classification has been performed on newspaper articles and has attained classification reports and accuracy. For conducting the research, getting started from selecting the dataset to preparing a pre-processed DataFrame for applying the ML algorithms and recording the results, the following steps that have been depicted in Figure 1, have been followed.

3.1 Dataset Collection

This research uses a newspaper article dataset of a Bangladesh-based newspaper named “Prothom Alo” that was derived by web scraping. The dataset contained almost 400K articles and it was in “dataset.json” file format. JSON stands for JavaScript Object Notation which is a type of dictionary in Python programming language structure. The following TABLE I represents one example of the dataset:

TABLE I. DATA SAMPLE OF SİNGLE DATASET

category	technology
content	গুগলের মোবাইল অপারেটিং সিস্টেম অ্যান্ড্রয়েডের বড় ধরনের ত্রুটির খোঁজ পেয়েছেন বলে দাবি করেছেন যুক্তরাষ্ট্রের প্রযুক্তি নিরাপত্তা গবেষণা প্রতিষ্ঠান ব্লুবক্সের গবেষকেরা। গবেষকেরা বলছেন, অ্যান্ড্রয়েডের দুর্বলতা বা বাগ হিসেবে একটি ‘মাস্টার কী’-এর খোঁজ পেয়েছেন তাঁরা। এ সফটওয়্যার ত্রুটির কারণে সাইবার অপরাধীরা অ্যান্ড্রয়েড স্মার্টফোন থেকে তথ্য চুরি করতে পারে। এক খবরে এ তথ্য জানিয়েছে বিবিসি অনলাইন।\xa0গবেষকেদের দাবি, অ্যান্ড্রয়েড সফটওয়্যারের এই ত্রুটি ২০০৯ সালের পর থেকে উন্মুক্ত সব সংস্করণেই রয়েছে। ক্রিপটোগ্রাফিক ভেরিফিকেশন প্রক্রিয়ার দুর্বলতার কারণেই অ্যান্ড্রয়েডে অতিরিক্ত ম্যালওয়্যারের আক্রমণ দেখা যায়। ক্রিপটোগ্রাফিক ভেরিফিকেশন প্রক্রিয়ার মাধ্যমে বিভিন্ন অ্যাপ্লিকেশন পরীক্ষা করে দেখা হয়। \xa0চলতি বছরের আগস্ট মাসে অনুষ্ঠিতব্য ব্ল্যাক হ্যাক হ্যাকারস সম্মেলনে অ্যান্ড্রয়েডের এ ত্রুটির বিস্তারিত জানানোর কথা জানিয়েছেন গবেষকেরা।\xa0অ্যান্ড্রয়েডের সফটওয়্যার ত্রুটির বিষয়ে গুগল কর্তৃপক্ষ আনুষ্ঠানিকভাবে কোনো মন্তব্য করেনি।

3.1.1 Dataset Construction

The dataset was collected from the newspaper's website [16] using web scraping and HTML trees. The dataset was created using the "beautifulsoup4" library and ten parameters were defined for each article and the distribution of classes for the dataset is depicted in TABLE II. The content of each article was too large to handle, so ten sentences per article were sliced as the "content" of the data. A loop function was implemented to slice the articles until the regular expression of a full stop in the Bangla language (|) came ten times. The first ten sentences of each article were fetched from the website, and the data was extracted in JSON format, which was then appended to a Python dictionary. Different links for different news categories were inserted, and corresponding data was collected.

TABLE II. DATA CLASSIFICATION

Dataset Category	**Number of Contents**	**Dataset Category**	**Number of Contents**
Bangladesh	5210	education	342
sports	1216	we-are	296
technology	460	onnoalo	139
entertainment	707	pachmisheli	87
international economy	804	askeditor	110
lifestyle	390	roshalo	46
opinion	190	others	2

3.2 Data-Preprocessing

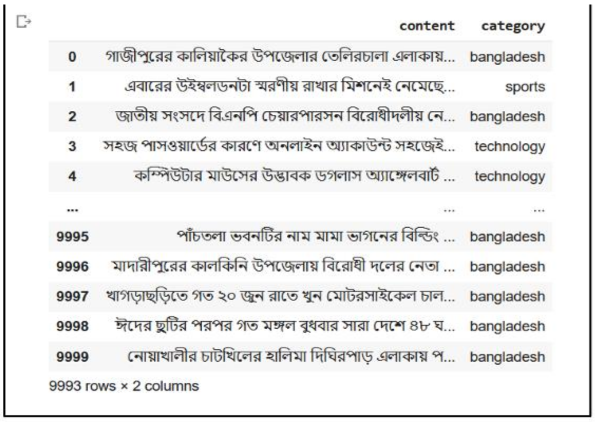
From figure 2, the pre-processing section can be observed which focuses on preparing a final data frame after removing HTML tags, stop words, non-Bangla word removal, null and duplicate data removal. An example of before and after pre-processing steps has shown in figure 3 & figure 4. The dataset used was derived from a website and contained HTML tags. To ensure accurate results, the tags were removed using a user-defined function and "lxlm" library. The raw dataset contained English words and other words, but only Bangla words were used. The library function "bnlp-toolkit" was used to import stop words such as **“কাছে, বা, এ, হয়ে”** and punctuations from the dataset.

Fig. 2. Data Pre-processing Flowchart

Fig. 3. HTML tag removal.

Fig. 4. Non-Bangla word removal.

The final data frame was generated by generating the "content" and "category" from the raw dataset. The first 10,000 data from the raw dataset were used to build the final data frame. Duplicate and null values were dropped using "df.dropna()" and "data.drop_duplicates()", resulting in a final count of [9993,2]. After All these pre-processing steps the final data was prepared for further experiment and an overview of the pre-processed data can be visualized from figure 5:



	content	category
0	গাজীপুরের কালিয়াকৈর উপজেলার তেলিরচালা এলাকায়...	bangladesh
1	এবারের উইম্বলডনটা স্মরণীয় রাখার মিশনেই নেমেছে...	sports
2	জাতীয় সংসদে বিএনপি চেয়ারপারসন বিরোধীদলীয় নে...	bangladesh
3	সহজ পাসওয়ার্ডের কারণে অনলাইন অ্যাকাউন্ট সহজেই...	technology
4	কম্পিউটার মাউসের উদ্ভাবক ডগলাস অ্যাঙ্গেলবার্ট ...	technology
...	...	...
9995	পাঁচতলা ভবনটির নাম মামা ভাগনের বিল্ডিং ...	bangladesh
9996	মাদারীপুরের কালকিনি উপজেলায় বিরোধী দলের নেতা ...	bangladesh
9997	খাগড়াছড়িতে গত ২০ জুন রাতে খুন মোটরসাইকেল চাল...	bangladesh
9998	ঈদের ছুটির পরপর গত মঙ্গল বুধবার সারা দেশে ৪৮ ঘ...	bangladesh
9999	নোয়াখালীর চাটখিলের হালিমা দিঘিরপাড় এলাকায় প...	bangladesh

9993 rows × 2 columns

Fig. 5. Overview of the Final Dataset after pre-processing.

3.3 Feature Extraction from Raw Data

Features must be extracted from the pre-processed dataset for text classification. Feature extraction means the conversion of text data or raw data into a numerical feature without changing the raw data's information. For this research, both the transformer-based embedding model which is BERT and the classical embedding method which is TF-IDF will be used for feature extraction [7].

3.3.1 Feature Generation Using TF-IDF

TF-IDF is a natural language processing algorithm that counts the frequency of words and their relevance to a given sentence. It consists of two parts: TF (Term Frequency) and IDF (Inverse Document Frequency) [14, 15]. TF works with a single term or word to determine how many times it has occurred in each document, while IDF measures the ratio of the frequency of that word to the total number of unique words in that document [8].

$$TF = \frac{\text{How many times the term has appeared in the document}}{\text{The total number of terms in the document}} \quad (1)$$

$$IDF = log(\frac{Number\ of\ \ document\ in\ a\ set.}{Number\ of\ \ documents\ that\ contain\ common\ term.}) \quad (2)$$

Inverse Document Frequency (TF-IDF) is an effective algorithm of NLP that employs Inverse Document Frequency (IDF) to classify texts and grasp their contents. It produces a matrix using sentences and words as characteristics, with the frequency of terms in columns. The TF matrix and IDF values are then multiplied. For feature extraction from articles, TF-IDF has been employed, creating 768 features for each content. This NLP approach is essential in interpreting the meanings of words in texts [9].

3.3.2 Feature Generation Using BERT

Bi-Directional Encoder Representations from a Transformer (BERT) is a language modeling technique that uses a transformer to analyze the relationships between words in a sentence. BERT consists of an encoder and a decoder, with the encoder reading and understanding the input text and the decoder making predictions. The following Figure 6. shows the embedding vector for one content of the dataset:

	feature1	feature2	feature3	feature4	feature764	feature765	feature766	feature767	feature768	category
0	0.701824	-1.160793	0.581013	0.309706	-0.380707	-0.707724	-0.090359	-0.749308	-0.144592	bangladesh
1	1.262319	-0.733864	0.504010	0.713356	-0.087414	-0.395833	-0.367399	-0.404731	0.449505	sports
2	1.308423	-0.790594	0.640789	0.649578	0.035974	-0.307077	-0.374227	-0.746825	-0.000746	bangladesh
3	1.199291	-1.185360	0.479411	0.386313	-0.056032	-0.337746	-0.559703	-0.226776	-0.153954	technology
4	0.756022	-1.045442	0.147317	0.235032	0.138051	-0.603351	-0.499416	-0.463948	0.136910	technology
...	...	...	...	...	...	...	...	...	...	...
9995	0.663784	-1.111204	0.488732	0.471967	0.112401	-0.607963	-0.143953	-0.653307	-0.106992	bangladesh
9996	1.316657	-0.830455	0.172440	0.194753	0.148920	-0.616923	-0.119074	-0.623211	-0.215042	bangladesh
9997	0.668249	-1.105444	0.191984	0.417070	-0.091570	-0.332222	0.047416	-0.611955	0.065397	bangladesh
9998	1.008497	-1.001944	0.060816	0.420068	0.026608	-0.389677	0.145841	-0.368035	0.164920	bangladesh
9999	1.074936	-0.956681	-0.193320	0.622191	-0.082608	-0.834676	-0.329707	-0.537791	-0.011086	bangladesh

10000 rows × 769 columns

Fig. 6. Overview of the dataset with feature vectors generated from BERT.

This research focuses on the encoder part of BERT, using a bidirectional approach to read the input. It uses an embedding vector, which is 768-shaped, to embed each word in a sentence, ensuring similar words are in the same category. BERT includes pre-trained models like 'Bert-base-uncased' and 'Bangla-Bert-Base'. In this research, a feature vector of 758 dimensions is generated for each article content.

3.4 Up Sampling & Down Sampling

Up sampled through the utilization of the "RandomOverSampler" library function, the sampling process equalized the classes of the dataset, generating data for minority classes while maintaining the major class's limit as a reference. This synthetic or artificial creation of data points effectively balanced the initially imbalanced dataset, as demonstrated in Figure 7, where three imbalanced classes were transformed into a distribution with equal representation. The application of up-sampling resulted in all category counts reaching 5210, with the "Bangladesh" category being the highest. The category distribution of the dataset post-up-sampling is depicted in Figure 8.

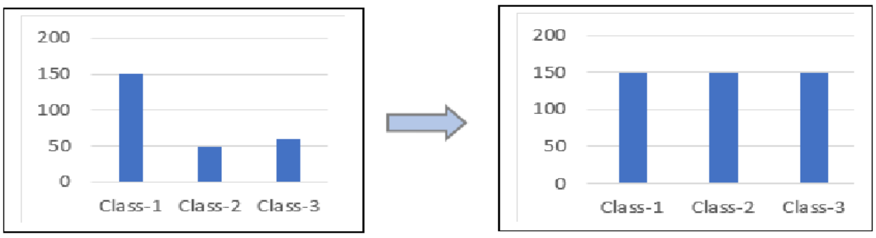


Fig. 7. Up-Samling process.

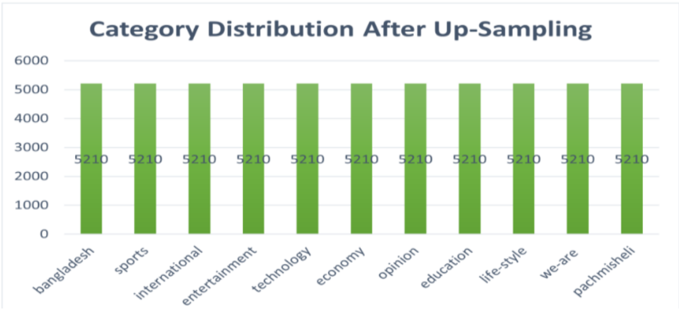


Fig. 8. Category distribution of the dataset after up-sampling.

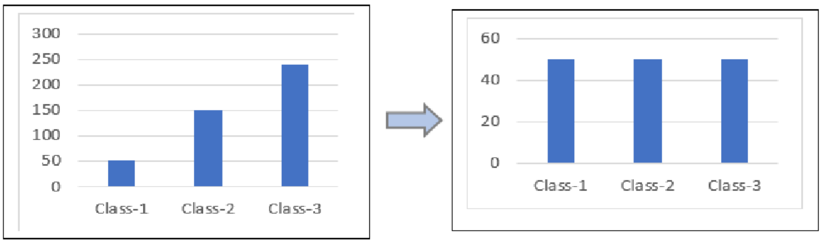


Fig. 9. Down-Sampling process.

The utilization of down sampling which can be seen in figure 9, a method similar to up-sampling, to reduce the count of a targeted class and balance the data in a dataset. In this process, data points above a specific range are eliminated, resulting in an evenly distributed representation of the targeted classes. The writer applied up-sampling in one research process to balance the data and, in another process, employed both up-sampling and down-sampling to achieve balance. The simultaneous application involved setting a limit of 500 counts, with categories exceeding this limit sampled using down-sampling, and those below sampled using up-sampling. The resulting distribution of categories is depicted in figure 10.

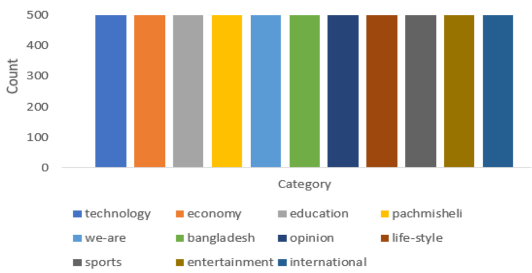


Fig. 10. The category count of the contents of the dataset after up down sampling.

3.5 IMPLEMENTATION OF ML ALGORITHMS

The research aims to compare text classification results from BERT and TF-IDF processes using machine learning algorithms on a pre-processed dataset. The cases that will be used for comparison have been stated in the following TABLE III:

TABLE III. EXPERIMENTAL CASES FOR THE RESEARCH

Text Classification using the Features Generated From BERT		**Text Classification using Feature Generated From TF-IDF**
1. Applying ML Algorithms on the imbalanced dataset (without sampling) **2.** Applying ML Algorithms on the balanced dataset (sampling the categories with up-sampling) **3.** Applying ML Algorithms on the balanced dataset (sampling the categories with both up & downsampling)	*VS*	**1.** Applying ML Algorithms on the imbalanced dataset (without sampling) **2.** Applying ML Algorithms on the balanced dataset (sampling the categories with up-sampling) **3.** Applying ML Algorithms on the balanced dataset (sampling the categories with both up & downsampling)

Four machine learning algorithms, a Decision Tree Algorithm, a Random Forest Algorithm, a Support Vector Classifier Algorithm, and a Logistic Regression Algorithm were used to classify the dataset. For applying the ML models, 67% of the data has been used for training and 33% of the data has been used for testing the approach [12, 13, 14]. A comparative analysis of the results will be achieved using some cases and cases are:

- Case-I: Implementing the ML Algorithms on the features without balancing the dataset.
- Case-II: Implementing the ML Algorithms on the features after balancing the dataset with up-sampling.
- Case-III: Implementing the ML Algorithms on the features after balancing the dataset with up-sampling & down-sampling and testing the model accuracy with unseen data.
- Case-IV: Implementing ML Algorithms on balanced dataset's most frequent features.

5 RESULT ANALYSIS & PERFORMANCE EVALUATION

The performance evaluation has been achieved based on the model's accuracy [15], which are described below:

Accuracy is a crucial metric for evaluating classification models, referring to the model's accurate prediction percentage on the classification.

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions} \tag{3}$$

TABLE IV. EXPEROMENTAL CASE STUDY'S ACCURACY RATE FOR DIFFERENT ML MODELS WITH BERT & TF-IDF

Case No.	Case I		Case II		Case III		Case IV
Case Title	Raw dataset		Up-sampling		Up-sampling & down-sampling		Balanced dataset's most frequent Items
Feature Extrator / **ML Models**	TF - IDF	BERT	TF - IDF	BERT	TF - IDF	BERT	BERT
Decision Tree	69.89	61.52	97.96	97.56	70.52	64.53	63.53
Random Forest Classifier	78.99	74.27	99.80	99.76	86.36	84.38	81.43
Support Vector Machine Classifier	86.33	85.39	98.46	99.29	85.77	87.53	83.47
Logistic Regression	83.63	86.70	95.44	97.82	81.48	87.96	82.53

For Case I, the targeted ML models have been applied on the raw features without any manipulation of the features and it can be visualized from figure 11 that the BERT model and TF-IDF was used as a feature extractor in the study. The logistic regression model achieved the highest accuracy rate of 86.70% when transformer-based feature extractor BERT was used. This model is best for multiclass classification problems, where linearly separable data is used. The decision tree classifier model had the least accuracy, with 61.52% accuracy. Random Forest Classifier and Support Vector Classifier achieved 74.27% and 85.39% accuracy, respectively. For the features that have been derived from TF-IDF model, Support Vector Classifier (SVC) had the highest accuracy rate of 86.33%, while Decision Tree Classifier (DT) had the least accuracy at 69.89%. The SVC model was also effective for multiclass-text classification, as it can tune feature parameters to achieve the best accuracy. The Random Forest Classifier and logistic regression models achieved 78.99% and 83.63% accuracy, respectively. However, as a tree-based classifier, it lags behind the logistic regression model. Logistic Regression is considered one of the best classifier models due to its efficiency in dealing with linearly separable multiclass datasets.

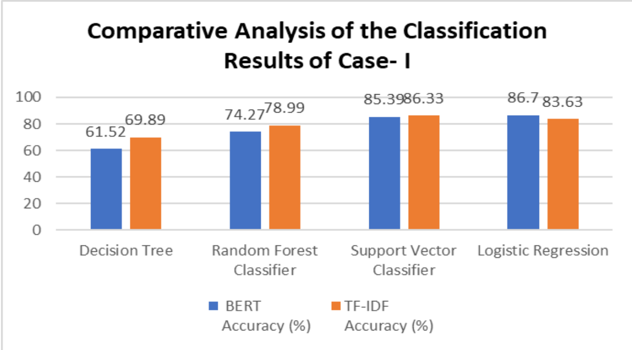


Fig. 11. Comparative Analysis between the Results of Classification of Case- I Using BERT & TF-IDF

For Case II, it can be observed from figure 12 that the study analyzed the accuracy of various machine learning algorithms for text classification using a dataset containing 5210 articles. The highest category was "Bangladesh", and all categories were up sampled to 5210.

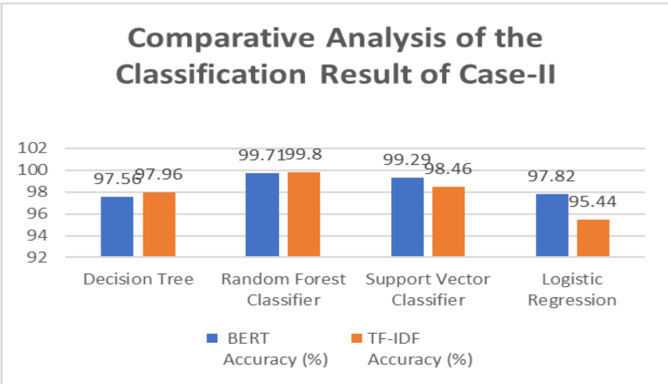


Fig. 12. Comparative Analysis between the Results of Classification of Case- II Using BERT & TF-IDF

For BERT- based feature extractor, the random forest classifier achieved the highest accuracy at 99.76%, based on its binary tree-based approach. The decision tree algorithm had the lowest accuracy at 97.56%, due to its binary class classification . The support vector classifier and logistic regression models achieved 99.29% and 97.82% accuracy, respectively. However, the random forest classifier was found to be the most accurate model for multiclass text classifications, with a 99.8% accuracy rate when the classical TF-IDF model was used as feature extractor. The logistic regression model had the lowest accuracy rate at 95.44%, due to the inaccurate parameter tuning during training of the dataset. The decision tree and support vector classifier models achieved moderate accuracy rates at 97.96% and 98.46%. The highest accuracy was achieved with a random forest classifier model, but it was overfitted due to dataset noise and duplicate data. Case II requires up-sampling and down-sampling to balance the dataset and achieve realistic accuracy.

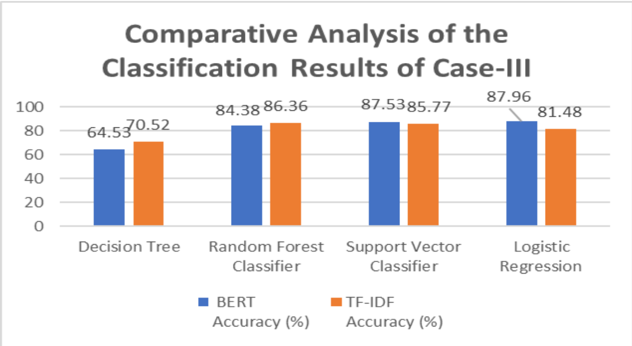


Fig. 13. Comparative Analysis between the Results of Classification of Case- III Using BERT & TF-IDF

The outcome of the Case III can be seen in figure 13, which focuses on implementing ML algorithms on features after balancing the dataset with up-sampling and down-sampling methods. The dataset consists of 5.5K articles with their corresponding sampled features and categories. The model accuracy is tested using unseen data. For BERT as feature extractor, the logistic regression model achieved the highest accuracy at 87.98%, due to its efficiency for linearly separable outcomes and data with less multi-collinearity. Due to small differences in class manipulation, the decision tree algorithm provided the lowest accuracy at 64.54%. Other machine learning models, such as random forest and support vector classifier, achieved 84.38% and 87.53% accuracy, respectively. The support vector classifier (SVC) model achieved a very good accuracy of 87.53%. The random forest classifier, a modified version of the decision tree model, achieved the highest accuracy at 86.36%. The support vector classifier and logistic regression models achieved 85.77% and 81.48% accuracy, respectively. These models lag behind the best-fitted model, possibly due to the features generated using the TF-IDF model and the information loss and duplicity of the data. The analysis of text features using BERT and TF-IDF showed the logistic regression model achieved the highest accuracy at 87.98%, outperforming the conventional model. The BERT model proved more efficient for classifying articles, and further experimental analysis is planned.

For Case IV, the transformer-based embedding method (BERT) is found to be effective for feature extraction in transforming news articles into their corresponding features. The most frequent 256 terms from the 5500 contents were extracted, and 768 features were generated. The Support Vector Machine Classifier model was found to have the highest accuracy of 83.47%, converting multiclass classification problems to binary ones and classifying using the hyper-plane. The random forest classifier model and logistic regression model also showed moderate accuracy rates, with the logistic regression model achieving almost 83% accuracy.

The overall experimental results have been represented in TABLE IV, where it can be perceived that the research analyzed four different cases, each with corresponding classification results. The first case dealt with an imbalanced dataset, where the categorizes were not sampled, resulting in an accuracy of 86.7%. The second case involved balancing the dataset with up-sampling categories, resulting in overfitting, and generating duplicate data. The third case involved a balanced dataset, where the logistic regression model achieved almost 88% accuracy when the feature extractor was the BERT embedding model. These results suggest that multiclass text classification works better with features generated using the BERT rather than the conventional TF-IDF model. The last case was designed for feature tuning with the transformer-based feature extractor, using only the most frequent 256 terms and articles with these terms for extracting features. The transformer-based embedding method generated features with the most frequent and relevant features, achieving an accuracy of almost 84%.

5 CONCLUSION

Text classification is an evolving sector of Natural Language Processing (NLP) that helps users find targeted content from websites or tags from large documents. This research aims to classify Bangla newspaper articles into relevant classes and determine if the BERT embeddings model or TF-IDF-based model works better for feature extraction from the newspaper's articles. Three cases were analyzed for this purpose: first, focusing on machine learning models with raw features, second, balancing the dataset with up-sampling on categories, and third, applying both up-sampling and down-sampling to train machine learning models. The last case proved the most efficient, with an accuracy of 88% with the logistic regression model. The BERT embeddings model was found to be the better feature extractor for text classification. The main contribution of this research is the construction of the dataset, handling imbalanced data through class manipulation, and performing feature engineering by fine-tuning the transformer-based embeddings. The results show that the BERT embedding model proved better than the conventional TF-IDF-based model for feature extraction. There were enormous challenges that has been dealt with while conducting the research as such the dataset was a bit larger hence it took a long time to execute necessary program, dataset construction was also a challenge as it was extracted from a newspaper website where real time news is being recorded on daily basis and so it was difficult to segment and train up the models. The advantage of BERT is that it can understand the contextual meaning of any language and provides a fixed width vector and the advantage of TF-IDF model is that it generates vector for any word by word for any language. On the contrary, the disadvantage of BERT is that it needs a pretrained model for each language and the TF-IDF models limitation is that the features extracted for each text of any dataset contains different length. This research can be extendedd for implementing news news-recommended system with transformer-based feature extractor and also a broad comparative analysis can be done involving some deep learning models as well.

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